

Defying the flat-Earth myth: Soil organic carbon monitoring in topographically complex temperate pasture landscapes

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ABSTRACT

Soil sampling design to capture the spatiotemporal variability of soil organic carbon (SOC) stocks in complex landscapes is challenged by the influence of topographical features on primary production, animal behaviour, and carbon and nutrient return to grasslands. Here we explore a range of approaches for streamlining the sampling process using measures of SOC stock data (2003, 2020, 2021) from a long-term hill-country fertilizer experiment. The stepwise procedure included a) evaluation of sampling efficiency between stratified and random soil sampling designs to capture spatial variability of SOC stocks, b) semi-parametric regression analysis to evaluate the representativeness of certain strata (e.g. medium slopes), c) exploring the appropriateness of using shallow SOC stock estimates to represent the full sampling depth, and d) the use of a process-based model to determine expected SOC stock changes and temporal variability to estimate the number of representative soil samples required to achieve a significant minimum change. Our analyses suggest that a) the stratified sampling design was more efficient reflected in the number of samples required to identify 5 % deviation in SOC stocks, b) estimates obtained on the medium slope were a reasonable approximation of the SOC stocks and its spatial variation across the farmlets, c) shallow (0–75 mm) SOC stock estimates from an adjusted regression model offered an accurate inference of its vertical distribution (0–300 mm), and d) the Grass-NEXT model provided insights on where more intensive sampling is required to detect a significant minimum change in SOC stocks over time. Tailored, stratified soil sampling and context-specific strategies, such as using representative strata and shallow sampling as effective steps for estimating SOC stocks and addressing data gaps, are crucial for simplifying monitoring and minimizing the cost of on-farm soil sampling designs in topographically complex landscapes covered by long-term pastures.

1. Introduction

Grasslands cover 20–40 % of the world's land and over two-thirds of the agricultural area (Reynolds, 2019). Defined as land on which vegetation is composed of perennial or self-seeding annual forage species (Allen et al., 2011), grasslands store 20 % of global soil carbon stocks (SOC) (up to 343 Pg SOC to 1 m depth), with an annual SOC sequestration of 0.5 Pg C (Lorenz and Lal, 2018). While small changes in SOC stocks can contribute to mitigate global warming (Stockmann et al., 2013), the detection of slight SOC variations under perennial pastures with high SOC stocks remains challenging in a diverse range of soils and

landscapes (Smith et al., 2020).

Worldwide, grasslands exist on both flat and topographical complex landscapes (Dixon et al., 2014). While methods for quantifying SOC spatial variability are well documented, FAO guidelines or national inventories (FAO, 2019; van der Weerden et al., 2020) offer limited advice for SOC sampling in farm landscapes with intricate micro-topographies. Accurate spatial characterization is crucial for SOC monitoring (de Gruijter et al., 2016; Mudge et al., 2020). Poor stratification and large uncertainty often demand many samples to reliably capture spatial SOC distribution (Malone et al., 2018). Topographical features in New Zealand hill pastures (more than 50 % of the country's grasslands), such as

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slopes over 15° and aspect (the direction a slope faces), significantly impact SOC storage by influencing water dynamics (Kemp and López, 2016; Zhong et al., 2016; Wiesmeier et al., 2019), grazing behaviour (López et al., 2003), and carbon and nutrient distribution (Bilotto et al., 2022). The long-term grazing experiment at Ballantrae Hill Country Research Station (Mackay et al., 2021a) offers a comprehensive dataset that details how topography, fertilizer, and grazing regimes affect SOC distribution in complex landscapes. Here, we hypothesize that stratified soil sampling designs using topographical features, are more efficient than random sampling in representing SOC distribution in such landscapes.

Soil samples collected from transects on medium slopes (12–25°) and shallow depths (0–75 mm) provide a good indication of SOC concentration (Mackay et al., 2021b) and biological activity (Schon et al., 2022). Monitoring soil fertility at these shallow depths to inform nutrient management in New Zealand pastoral systems is routine along transects on medium slopes of major land units (Morton et al., 2000; Dawson and Knowles, 2018). Surprisingly, few attempts have integrated SOC stock monitoring with nutrient and resource management in farm plans. Kelliher et al. (2012) explored the relationship between shallow and deep SOC distribution in a long-term pasture in Winchmore, New Zealand, suggesting that SOC stocks from shallow samples could be extrapolated to greater depths cost-effectively. This method could help to define representative areas for deeper sampling where greater accuracy is required (e.g., for SOC stock reporting). Given the difficulty, labour intensity, and danger of soil sampling in complex landscapes, we hypothesize that frequently monitoring representative areas on medium slopes with shallow samples (0–75 mm) could significantly and accurately facilitate soil sampling in hard-to-reach places.

Depending on climatic effects, soil type and management practices, SOC loss or accumulation after land use change is a slow process, taking years or even decades to reach a new equilibrium level (Schipper et al., 2017; Li et al., 2018). Soil sampling designs must account for spatial and temporal variability to ensure an accurate representation of SOC stocks and SOC changes over time. The assessment of the temporal variability of SOC stocks often requires multiple sampling rounds, typically every 3–5 years, which can be challenging for farmers to maintain (Allen et al., 2010; FAO, 2019). Alternatively, where data is lacking, SOC stocks can be preliminarily assessed using process-based models that evaluate C cycle dynamics across the soil–plant–grazing ruminant continuum (Smith, 2004; Smith et al., 2020). However, few process-based models consider the influence of topographical features on SOC stocks in grazing systems (Bilotto et al., 2022). Here, we aim to test how SOC temporal variability, simulated by a calibrated process-based model designed for soil P, C, and N dynamics, can be combined with sampled spatial variability to optimize sampling strategies.

The main aims of this work were to i) compare the sampling efficiency between a stratified and a random design to detect differences in SOC stocks, ii) identify representative areas to quantify SOC stocks, iii) test if shallow samples (0–75 mm) could be adapted to provide a first approximation of the full SOC profile (0–300 mm), and iv) explore the role of Grass-NEXT (Bilotto et al., 2022), as a locally calibrated process-based model, to assist in the soil sampling design and monitoring of SOC stocks in topographically complex landscapes. This paper builds on existing approaches to carbon and nutrient, and natural resource management in grazing systems and on different approaches to improve and facilitate their inclusion in the farm plan and in reporting on the state of on-farm resources.

2. Materials and methods

2.1. The sequence of procedures

Facing the apparent paradox of minimising sample area and maximising the likelihood of detecting SOC temporal changes whilst minimising spatial variance, this study advances a sequence of steps for

designing soil sampling regimes to monitor SOC over space and time in topographically complex landscapes (Fig. 1). These steps were:

- Given the substantial impact on SOC dynamics, finer resolution and map scale is necessary to faithfully resemble major landforms, soil types and topographical features (slopes and aspects) (scale of mapping 1: <10,000 vs 1:50,000).
- Then, to compare the efficiency between soil sampling designs (*stratified* and *random*), we ran power analysis with data collected from a long-term study (Mackay et al., 2021a) based on the mean, variance and expected change of SOC stocks to calculate the number of samples required (*Sampling design* in section 2.4) (Fig. 2).
- Subsequently, to understand if the soil sampling and monitoring of medium slopes is appropriate as representative of SOC stocks in Hill Country farms, a geostatistical analysis and a comparison of the mean of SOC stocks at 300 mm soil depth from all three slope classes vs medium slopes, were developed (*Representativeness* in section 2.5).
- We optimized and validated the SOC vertical distribution equations calibrated by Kelliher et al. (2012) to extrapolate SOC stocks to a 300 mm depth and report SOC stocks when soil samples are collected at 75 mm depth. (*Vertical distribution* in section 2.6).
- Finally, the expected SOC stock changes and temporal variation regarding the topographical features were estimated using a process-based model (Bilotto et al., 2022) to determine the number of samples required (*Number of samples* in section 2.7).

The results from this study were discussed to gradually facilitate, advance, and incorporate the monitoring of SOC as part of a complete assessment of soil health, and provide an estimation of SOC stocks for the assessment of the condition of natural capital stocks and C accounting. The sequence of procedures and analysis *in series* are described in detail below.

2.2. Study site and soil sampling

The Ballantrae Hill Country Research Station is located in southern Hawke's Bay (40°18'S 175°50'E), 300 m.a.s.l. with mean long-term annual air temperature and rainfall of 12.8 °C and 1270 mm, respectively. The soils are classified as Andosols and Luvisols with a silt-loam texture (Lambert et al., 2000). The farmlets included in this study are part of a larger study described in Lambert et al. (2000). Three self-contained farmlets were examined (NF, LF, HF) that have been under different fertiliser and sheep grazing regimes since 1980: NF (9.7 ha): no single superphosphate (SSP) applied; LF (8.1 ha): 125 kg SSP ha⁻¹ yr⁻¹; HF (6.8 ha): 375 kg SSP ha⁻¹ yr⁻¹. Farmlets were stocked with sheep to maintain similar grazing pressure across farmlets i.e., similar stock units (SU) per unit of pasture production. Corresponding annual mean stocking rates have been about 6.0, 10.6, and 16.1 SU ha⁻¹ (1 SU=550 kg DM).

On each of the three farmlets, soils were sampled to a 300 mm depth (0–75, 75–150 and 150–300 mm) in 2003, 2020 and 2021, and to 0–75 and 75–150 mm in 2014. Within each farmlet, soil samples were taken from 18 permanently marked sites established in 1975 (3 slope classes × 3 aspect locations × 2 reps = 18 soil samples). To represent the topography of the farmlets, the permanent sites cover three slope classes [low slope (LS; 1–12°), medium slope (MS; 12–25°), high slope (HS; >25°)], on three different aspect locations grouped relative to the true north [East (E; 35–155°), Southwest (SW; 155–275°), Northwest (NW; 275–35°)] (Table 1). Each slope class and aspect location were replicated twice within each farmlet. For more information on soil sampling procedures and sample analysis see Mackay et al. (2021a).

2.3. Estimation of soil organic carbon stocks

To quantify total SOC stocks at each layer depth (expressed as Mg C/ha), soil C concentration (OC; % or mg C/g soil) and bulk density (BD; g

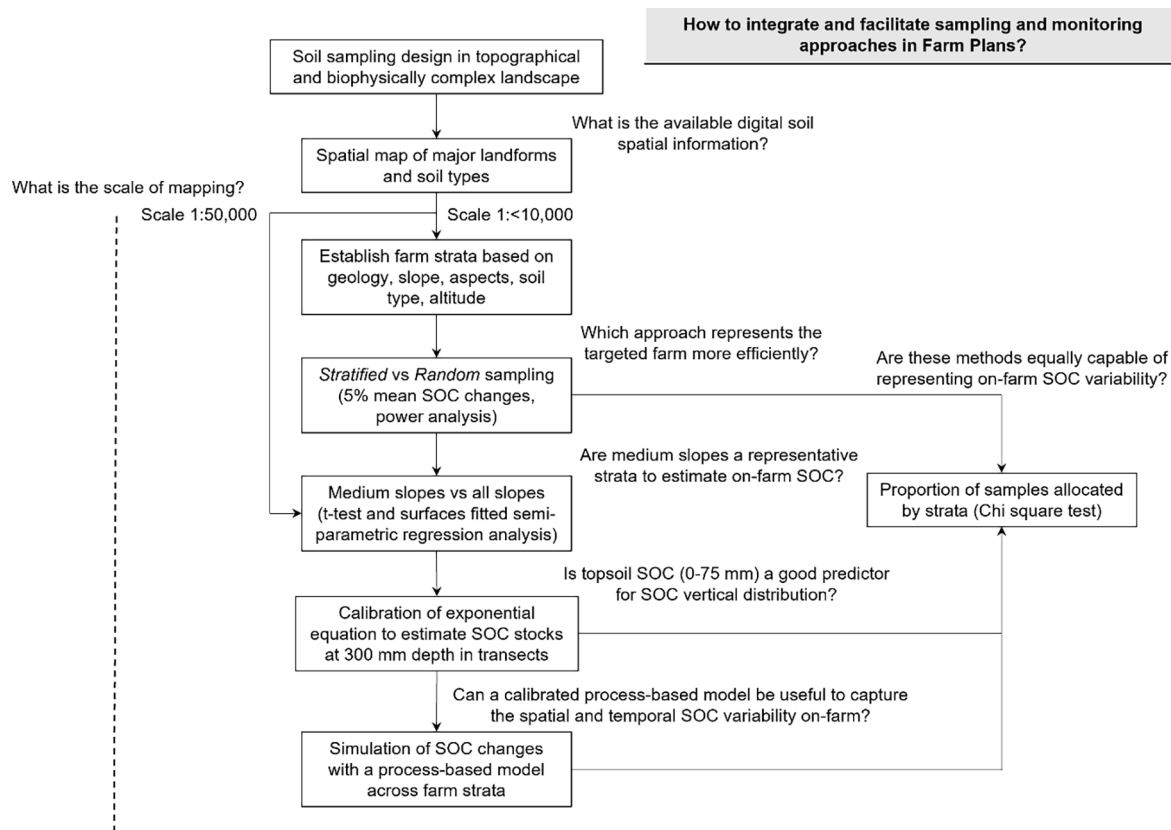


Fig. 1. Study overview and sequence of procedures integrated *in series* for soil sampling design in complex landscapes.

cm^{-3}) estimates were used according to the following equation:

$$\text{SOC}_{i,\text{stock}}(\text{MgCha}^{-1}) = \text{OC}_i \times \text{BD}_i \times (1 - \nu G_i) \times t_i \times 0.1$$

where, SOC_i = soil organic carbon stock (in Mg C/ha) at each layer depth i (1 Mg = 1000 kg), OC_i = organic carbon (mg C/g soil) of the fine soil fraction (<2 mm) in the depth increment i , BD_i = bulk density, the mass of fine soil per volume of fine soil of the depth increment i (g fine soil cm^{-3} fine soil) = dry soil mass [g] – coarse mineral fragment mass [g]/(soil sample volume [cm^{-3}] – coarse mineral fragment volume [cm^{-3}]), νG_i = the volumetric coarse fragment content of the depth increment i , t_i = thickness (depth, in cm), at each depth layer i , = conversion factor to convert mg C cm^{-2} to Mg/ha .

2.4. Sampling design to characterise hill country landscapes

To build a theoretical curve of samples required for different power levels and compare the efficiency of stratified vs random sampling, the SOC stocks were converted to log10 getting a log-normal distribution (Allen et al., 2010). Following, a statistical power analysis was performed to detect a difference within 5 % of the sample mean and considering their given standard deviation, with 95 % confidence (two-tailed test). Here, the power level or statistical power measures the likelihood that a test will correctly identify a true positive, expressed as a value between 0 and 1. A higher power level indicates a reduced risk of a Type II error, which occurs when a test fails to reject a false null hypothesis. Soil samples were collected from the HF farmlot to a depth of 300 mm in the spring of 2021 using a random and a stratified static synchronous sampling design (herein *stratified* sampling design, using the permanent sampling sites of the original long-term experiment layout). The stratified design can be a simple method, cheap and easy to understand for monitoring changes in SOC stocks at the farm scale (Mudge et al., 2020). Here, eighteen composite samples were collected

from permanently marked, independent sites with a suitable representation of each topographic feature within the farmlot (3 slopes classes x 3 aspects locations x reps = 18 samples). Then, we compared the *stratified* design with SOC stocks measured from samples collected from 18 locations randomly selected from a grid (herein *random* sampling design). A detailed sequence of maps, describing soil sampling designs (*stratified* and *random*), soil sample allocation, and topographical features, is presented in [Supplementary Material \(Figure S1\)](#).

2.5. Representativeness of medium slopes for soil sampling of SOC stocks

We explored whether data collected for SOC stocks from medium slopes (MS; 12 – 25°) provide a good representation of SOC in hill country. The SOC stocks data from the MS were compared to data collected from across all three slope classes. Homoscedasticity (homogeneity of variance) and normality were tested. The data collected for SOC concentration (%) and stocks across farmlots and topographical features comply with the parametric test requirements, then comparisons between the values of SOC stocks obtained across all slopes vs medium slopes were carried out by analysis of variance (ANOVA). Significance between groups ($p < 0.01$) was determined using Tukey's test. The distribution of the samples was explored in box plots (includes means, upper and lower quartiles, and outliers) and rainclouds (the point and line in the centre of each cloud represents its mean and 95 % CI).

A geospatial analysis was also conducted to assess the spatial distribution of SOC stocks and variability across the study area and provide confidence in any proposed sampling design. The SOC stocks were fitted by a smooth surface based on geospatial coordinates (Easting and Northing) with median slope indicator using a generalized additive model (Wood, 2017). In the model, a single common surface was estimated using all slope data, with a different surface estimated for medium slope data, the difference between these two surfaces was tested

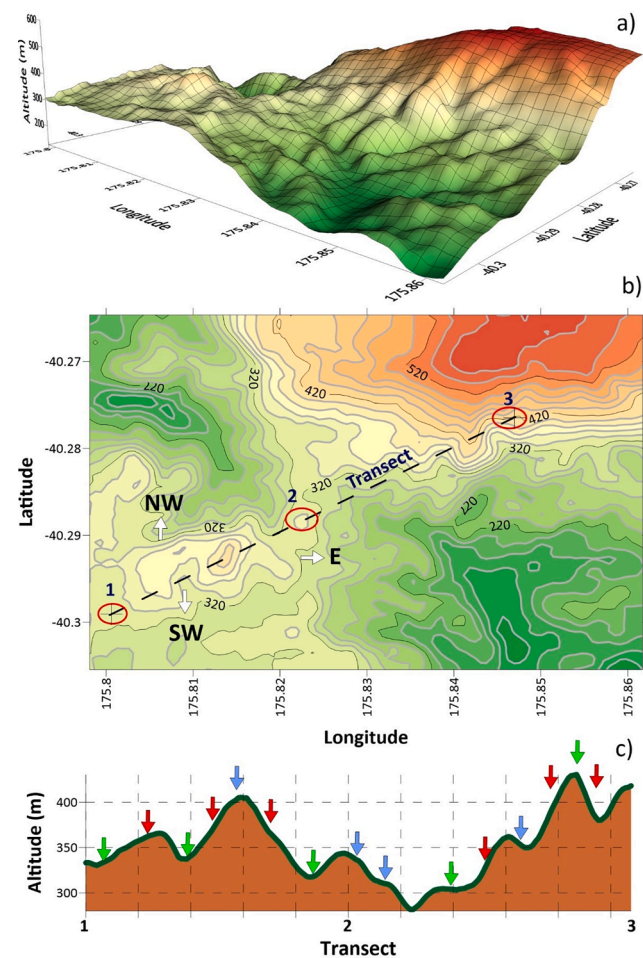


Fig. 2. Topographic representation of Hill Country landscapes in New Zealand (North Island). a) 3D surface digital elevation map. b) Contour and digital elevation map with aspects: East (E; 35–155°), Northwest (NW; 275–35°) and Southwest aspect (SW; 155–275°). c) Topographical profile: green, blue and red arrow depict low (0–12°), medium (12–25°) and high (>25°), respectively.

Table 1

Land area (%) in each slope and aspect combination within each farmlet. NF: no single superphosphate (SSP) applied; LF: 125 kg SSP ha⁻¹ yr⁻¹; HF: 375 kg SSP ha⁻¹ yr⁻¹, since 1980. Three categories of slope and aspect were used to describe the topography of the farmlets: low (LS; 0–12°), medium (MS; 12–25°) and high slope (HS; >25°) on different aspects grouped relative to the true north as East (E; 35–155°), Southwest (SW; 155–275°) and Northwest (NW; 275–35°) aspects.

Farmlet	Slope	Aspect		
		E	SW	NW
Land area (%)				
NF	LS	14	4	2
	MS	30	11	10
	HS	16	2	11
LF	LS	7	11	10
	MS	8	16	22
	HS	4	3	20
HF	LS	5	4	9
	MS	16	10	33
	HS	12	3	9

Adapted from Bilotto et al. (2022).

and are presented in a 3D-graph. The spatial analysis was developed through a fitted semi-parametric regression to the response SOC stocks to the fixed effects of ‘Easting’ and ‘Northing’ at 0–75, 75–150, and 150–300 mm depth in R (R Core Team, 2022). For inference of differences among fitted surfaces were produced for the significant result. All analyses were performed using R 4.21 and its packages ‘predictmeans’ (Luo et al., 2021).

2.6. Estimation of SOC vertical distribution

A regression analysis was performed to describe the relationships between SOC stocks to 75 and 300 mm depths. The SOC vertical distribution is defined by changes on soil properties and relationships with depth. The current analysis requires a continuous function of SOC stocks in depth, where corroborative measurements have been made on samples collected at discrete depth intervals (0–75, 75–150, 150–300 mm) across the farmlets and slope classes in 2003 and 2021. Soil organic carbon density (SOCD) was estimated using the following procedure:

$$\text{SOCD (kg C m}^{-3}\text{)} = \frac{\text{SOC}_i \times 100}{D_i}$$

where, SOC_i is the SOC in Mg C/ha at each layer depth, and D_i is the depth of each layer (e.g., D_i for the 150–300 mm depth layer is the difference between the upper and lower depth, D_i = 300–150 = 150 mm).

To estimate the SOCD for each slope and aspect combination, the exponential curve developed by Kelliher et al. (2012) to describe the vertical distribution of SOCD for Winchmore, New Zealand, was used. The curve was fitted by regression analysis with a nonlinear adjustment of the length scale and SOCD near the surface by the least square method in solver, minimizing the SOCD differences between observed and estimated data (Table S1). Then, the equations shown in Table S1 were used to extrapolate SOC stocks at the upper depth (0–75 mm) to lower depths (75–150 and 150–300 mm) in 2020 as recommended by FAO (2019). In addition, the observed and predicted SOC stocks across farmlets and topographical features (slopes and aspects) in 2003 and 2020 were compared to calculate SOC changes and the number of soil samples required for a new hypothetical sampling round.

2.7. Grass-NEXT model and its contribution to estimate the number of samples regarding the topographical features

Understanding the spatial patterns of soil C across hill country is a key element in the design of any soil C-stock monitoring regime and to estimate the spatial and temporal effects of topography and animal behaviour on soil C changes. The locally calibrated Grass-NEXT model (Bilotto et al., 2022) was developed using the simulation and modelling tool Insight Maker (Fortmann-Roe, 2014). A wide range of soil attributes are accounted for, along with the influence of topography on primary production, pasture composition and animal behaviour (including grazing, camping, and nutrient return), as well as how these affect the dynamics of the biogeochemical cycles of P, C and N soil stocks over time (Bilotto et al., 2022).

The initial conditions for SOC stocks and bulk density were assumed from Mackay et al. (2021b) extended to a 300 mm depth. Model parameters and equations used to calculate mean changes and variability (standard deviation) in SOC stocks to a depth of 300 mm are detailed in Bilotto et al. (2022). These parameters and equations are essential to determine the number of samples needed (or alternatively the number of years required) to detect a statistically SOC change. Here, we assumed a topographically stratified static synchronous SOC stocks monitoring design (Mudge et al., 2020) where the same sites are revisited in each monitoring round, since this requires the smallest number of samples to

detect a specified change in SOC stocks. For this section, we assumed 5 Mg C/ha to a paired sample, two tail, *t*-test power analysis with a significance level of 0.05 and a power of 0.8. The modelled SOC stock changes and its standard variation (temporal variability) between 2003 and 2020 across the farmlets in the Grass-NEXT model was essential to determine the number of samples required applying the minimum detectable difference (MDD) calculations (or alternatively the number of years required) to detect a statistically SOC change (Smith, 2004). To capture effects of topographical complex landscapes on sampling regimes, varying the proportion of slopes and aspects on topographical complex landscapes (see [Supplementary Material, Table S2](#)), we pooled both the spatial and temporal standard deviation of SOC stocks according to the following equation:

$$\text{Pooled SD} = \sqrt{\text{SD of spatial SOC stocks differences}^2 + \text{SD of temporal SOC stocks change}^2}$$

where SD is standard deviation.

2.8. Statistical analysis to compare the methodologies applied to quantify SOC changes and the expected proportion of samples allocated by slope and aspect

The Pearson's chi squared test and a contingency table was prepared to corroborate statistically significant differences between the observed, estimated and modelled frequencies of samples allocated by slope and farmlet (*p*-value < 0.05). All analyses were done with R 4.21.

3. Results

3.1. Theoretical efficiency of stratified and random soil sampling design to detect significant SOC stocks changes in hill country grasslands

To detect a difference of SOC stocks within 5 % of the sample mean and considering their standard deviation in the HF farmlet (transformed

to logarithm given that statistical power analysis requires a normal distribution), the stratified design was more efficient than the random soil sampling design (Fig. 3). This efficiency gain was reflected in the number of samples required to attain a predetermined statistical power. We found that the sample mean for the stratified soil sampling design was 2.04 [$\log_{10}$ (Mg SOC ha⁻¹)], and the standard error was 0.013 [$\log_{10}$ (Mg SOC ha⁻¹)] whilst the random design showed a sample mean of 2.02 [$\log_{10}$ (Mg SOC ha⁻¹)] and the standard error was 0.022 [$\log_{10}$ (Mg SOC ha⁻¹)]. When the samples were equally distributed (SED) or proportionally distributed (SPD) between slopes (based on the shared area from each slope) across the different farmlets (Figure S2), the cumulative probability of the mean SOC stocks was dominated by SPD in the LF and NF farmlets (~2 Mg SOC ha⁻¹) whilst the standard error was 0.2 Mg

SOC ha⁻¹ higher in the SED approach for the HF farmlet.

3.2. Soil sampling distribution and geospatial analysis of SOC by slope classes

Analysis of the soil data from the long-term phosphorus fertiliser and sheep grazing experiment consistently showed that sampling on a medium slope is a good representation of SOC concentration and stocks across all slope classes, depths and between years in hill country landscapes (Fig. 4). No significant differences were found in estimates of means, distributions and interquartile ranges of SOC stocks from soils sampled on the medium slope compared with all three slopes. Although the mean of SOC stocks in the full vertical profile were not different between medium and all slopes, the spatial analysis developed through a fitted semi-parametric regression to the response SOC stocks to the fixed effects of 'Easting' and 'Northing' revealed some extreme differences in the deepest layer (150–300 mm) (Fig. 4).

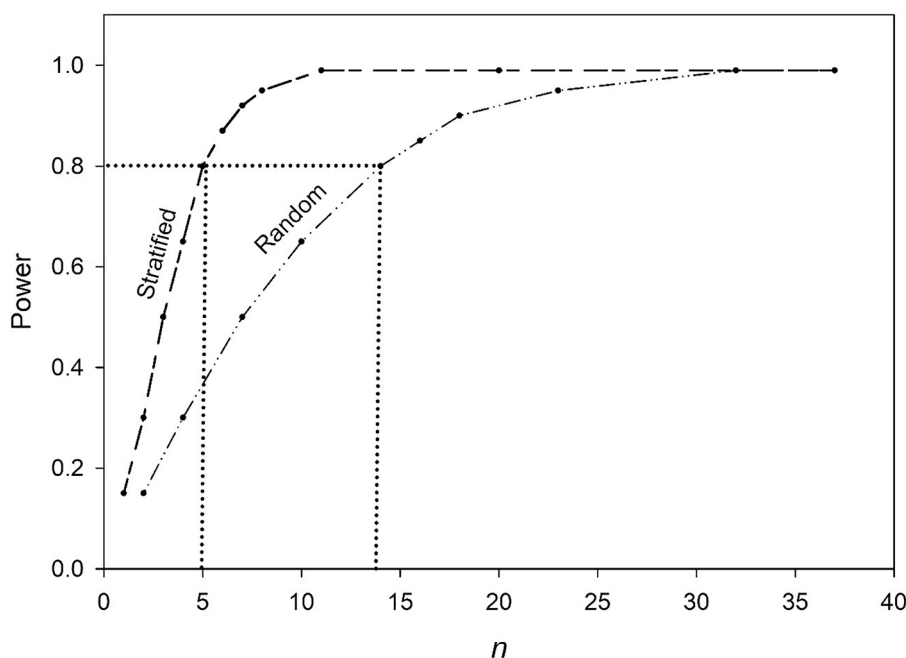


Fig. 3. Statistical power to detect a deviation within 5 % of the sample mean in the HF farmlet in 2021, with 95 % confidence (two-tailed test): dashed line = stratified sampling; dash-dotted line = random sampling.

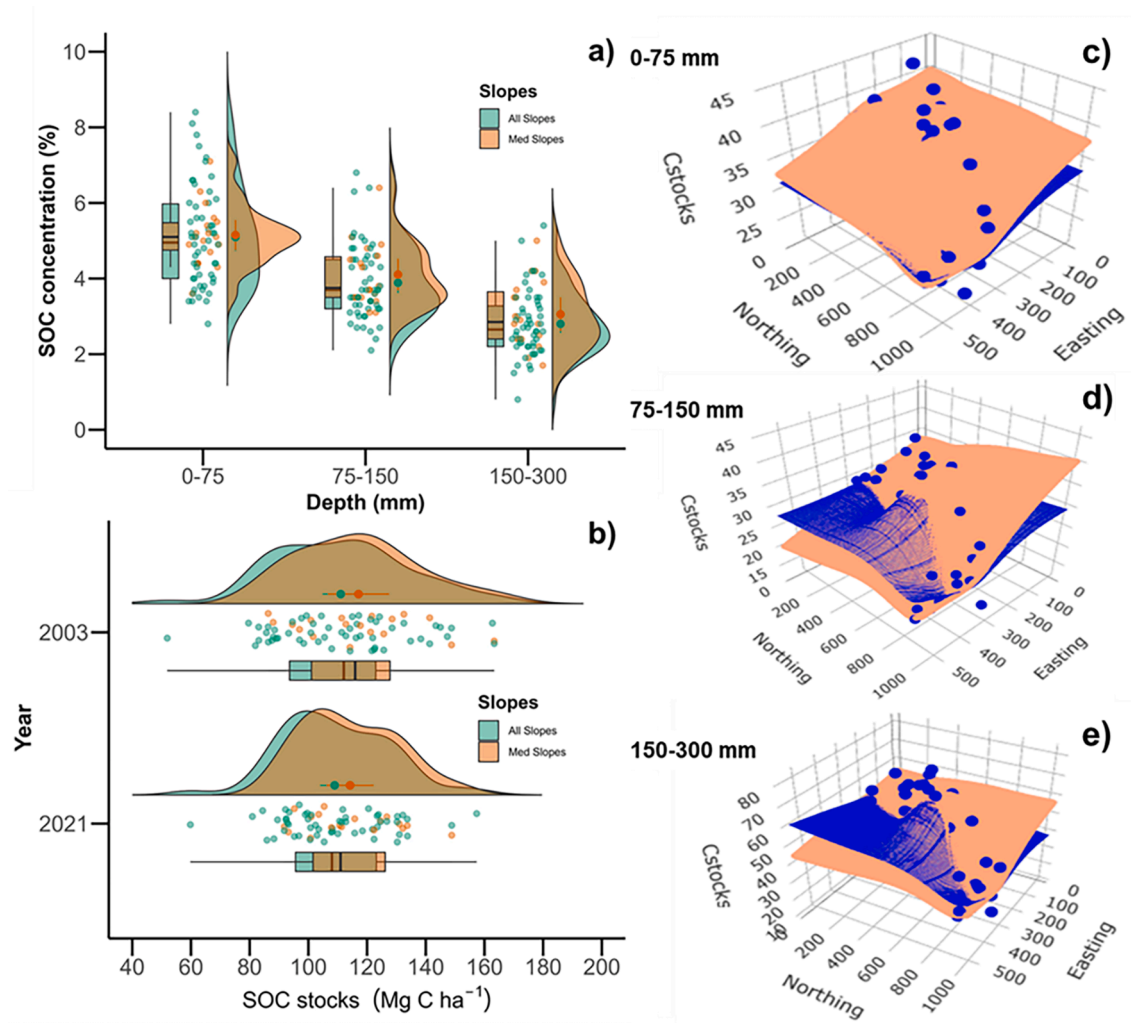


Fig. 4. Soil organic carbon (SOC) at 0–300 mm depth across all three slopes or when limited to medium slopes at the Ballantrae Hill Country Research Station. The *t*-test showed no significant differences ($p > 0.05$) of SOC concentration between medium slopes and all slopes across each individual depth (a) and between years (b). The mesh surfaces confirm these similarities by spatially overlapping SOC stocks between medium and all slopes across each individual depth (c, d, e). The point and line in the centre of each cloud represents its mean and 95 % CI. The rain represents individual data points. Box plots show min, max, 25th, 50th, and 75th percentiles.

3.3. Association between shallow SOC stocks and its vertical distribution across slope classes and farmlets

The equations calibrated and used to calculate the vertical distribution of SOC stocks gradually showed a higher SOCD in the NF, LF and HF farmlet, respectively (Fig. 5a–c). The NF farmlet exhibited significant differences of SOCD between HS and LS or MS at the 75–150 mm and 150–300 mm depth but no significant differences were observed between slopes at 0–75 mm depth (Fig. 5a). Overall, significant differences of SOCD were observed vertically in the full soil profile between in LS and MS with HS (Fig. 5b and c). The adjusted R^2 showed a significant moderate relationship of SOC stocks between 0–75 mm and 0–300 mm depth which was consistent between 2003 and 2021 sampling years (Fig. 5d and e). The observed SOC stocks in 2020 showed a strong and significant relationship with those estimated indirectly from the equation calibrated for SOCD calculation across slopes and farmlets (Fig. 5f).

3.4. Modelled temporal variation of SOC stocks across slopes classes and farmlets, sampling intensity or years to detect a significant change in SOC stocks

By using the distribution of SOC stocks for 2003 as a baseline, the

simulated SOC changes from the Grass-NEXT model showed a gradual reduction in SOC stocks in slope classes and aspect locations with high initial SOC stocks, but no change or slight gains were observed at sites with low initial stocks (Fig. 6). While the simulated SOC changes in the 10-year period were uniformly estimated across the HF farmlet (1–3 % change on initial SOC stocks), the NF and LF farmlets had tangible reductions in the SW and E aspects and increases in the LS from NW and E aspects (~5% change on initial SOC stocks), respectively. At an equal temporal variation, the number of samples required to detect a significant change varied inversely with the rate of change in SOC stocks. The NF and LF farmlets have taken 10–20 years to detect a 5 % of SOC change whilst the HF farmlet will most likely require 20–50 years to demonstrate the same rate of change.

3.5. Allocation of samples by slope class to detect a significant change in SOC stocks between the methods proposed and collected data

Regardless of the number of samples needed to detect a significant change in SOC stocks (Table S3), we found that the allocation of samples across the farmlets on different slopes (applying equations to estimate SOC stocks at depth from topsoil samples or using process-based simulation models) did not show significant differences (Fig. 7). Compared to

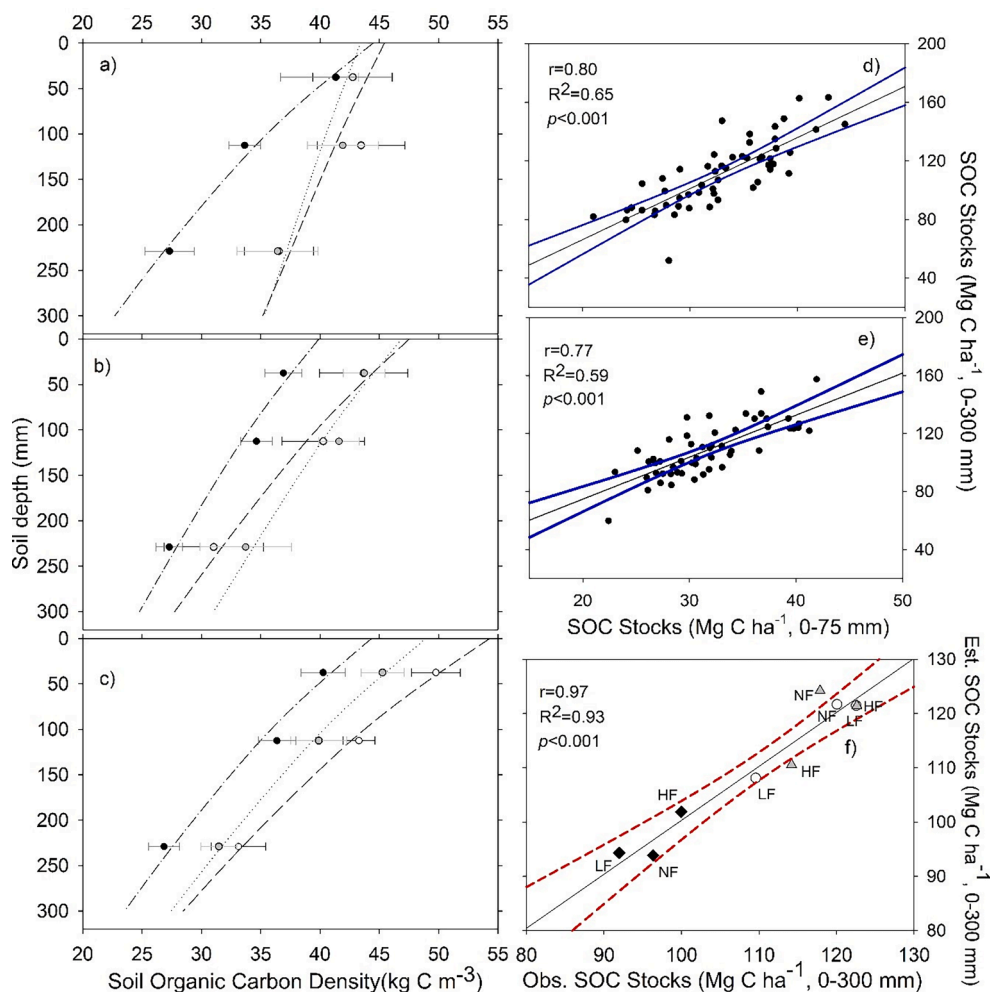


Fig. 5. Observed soil organic carbon density (SOCD; dots) and curves fitted by regression analysis with a nonlinear adjustment by the least square method in solver (lines) across slope class within each farmlet. a) NF (no SSP applied); b) LF (125 kg SSP ha⁻¹ yr⁻¹); c) HF (375 kg SSP ha⁻¹ yr⁻¹, since 1980). Open, grey and black dots represent SOCD observed at LS (Low Slope), MS (Medium Slope) and HS (High Slope), respectively. Dashed, dotted and dot-dashed lines represent SOCD curves calibrated at LS, MS and HS, respectively. Error bars represent standard error. Linear regression (solid line) between SOC stocks at 0–75 mm and 0–300 mm depth across farmlets in 2021 (d) and 2003 (e), respectively. Linear regression between observed and estimated SOC stocks at 0–300 mm depth across low, medium and high slope classes (open, grey and black points, respectively) and farmlets in 2020 performing the equations developed in Table S1 (f). Blue and red dashed lines represent 95 % confidence interval. r = correlation coefficient; R^2 = Coefficient of determination; $p < 0.05$ significant association between observed and estimated data.

the analysis of sampling data, both approaches have produced similar representations of sample allocation for the NF farmlet. However, the process-based model Grass-NEXT allocated a higher proportion of samples to HS in the LF farmlet, whereas the *adjusted Kelliher equations* allocated a higher proportion to HS in the HF farmlet (Fig. 7).

4. Discussion

4.1. Soil sampling strategies for SOC monitoring in topographically complex landscapes

Determining the true value of SOC stocks for a certain spatial unit is impossible, but reporting a mean value with its uncertainty is essential. Uncertainty, defined as the lack of knowledge of a true value (FAO, 2019), is shown by the standard error and sample size needed for a given statistical power. The chosen sampling strategy affects variance, which can be evaluated in advance. Identifying error sources in SOC stock estimation is crucial to minimize the minimum detectable difference (MDD) and sample size. The best strategy to reduce uncertainty involves design-based statistics, using random or stratified sampling. Here, to detect a difference in SOC stocks within 5 % of the sample mean, the

stratified design was more efficient than the random sampling design due to the lower variance detected (Fig. 3). This finding aligns with previous evidence, as simple random sampling can cause undesirably large portions of the study area to go unsurveyed by clustering samples in other areas (Allen et al., 2010; Shao et al., 2021).

As was observed by Mackay et al. (2021a), the spatial variability of SOC stocks in topographical complex landscapes (*hill country pastures*) depends on a variety of factors, including topography and soil type, and to a lesser extent, plant growth, climate and site management. Based on the results obtained by Mudge et al. (2020), we recommend a stratified static synchronous study design, where the samples are still collected randomly by strata. For subsequent monitoring rounds, the samples should be collected in fixed proximity (e.g., within a 5-meter radius), given that this design requires the least number of samples to detect changes in SOC stocks.

Often, the sample size allocation is an issue, and the most suitable design may vary according to how the homogenous units or strata were predefined (Shao et al., 2021). Although, the weighted average with samples proportionally distributed between slopes is less biased towards the middle value of SOC stocks and usually assigns an average value where most of the observations lie within (Figure S2), the weighted

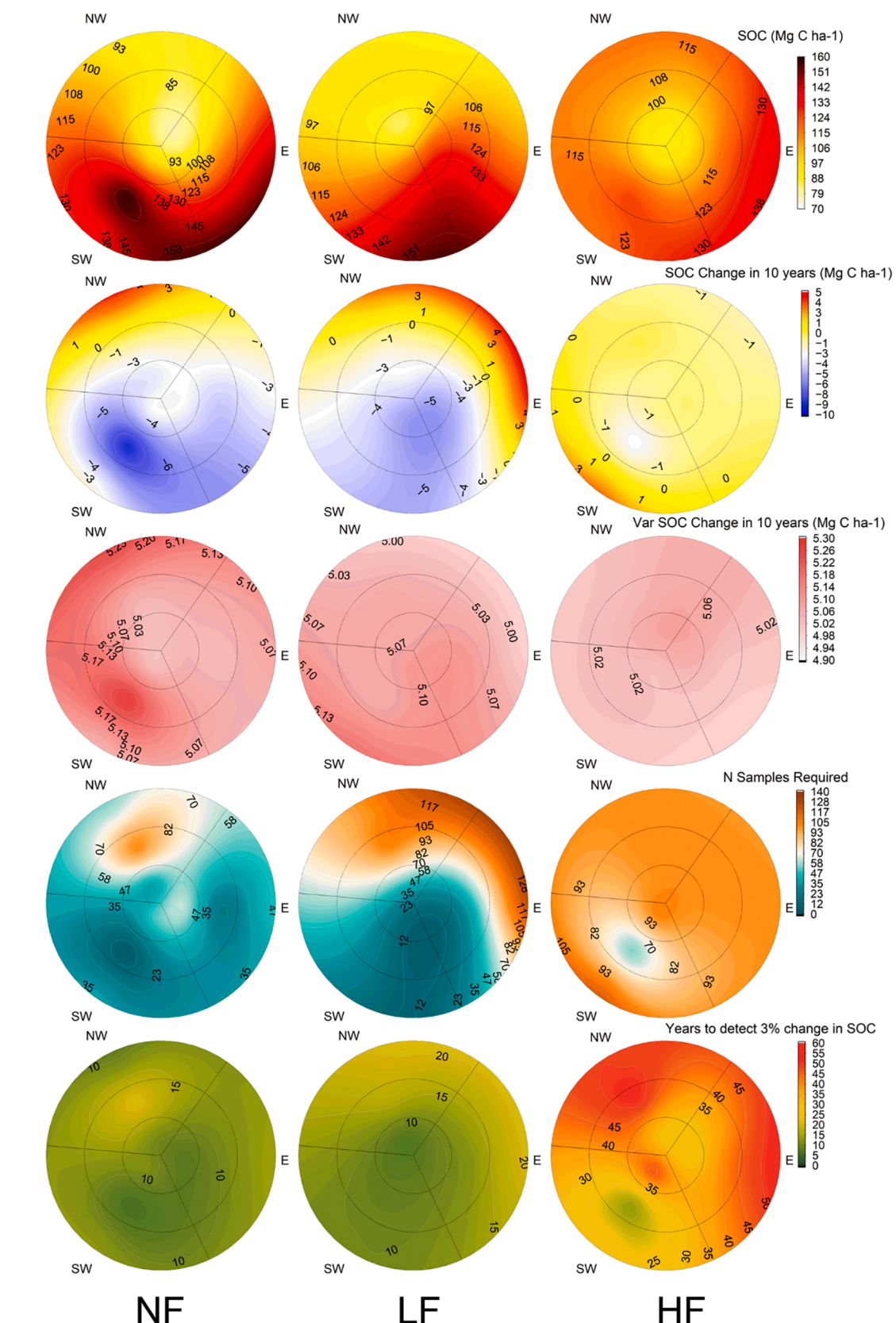


Fig. 6. Modelled SOC changes, standard deviation, number of samples required and years to detect a 3 % change in SOC stocks across farmlets from data collected in 2003 using the Grass-NEXT model (Bilotto et al., 2022). Farmlets: No fertility (NF; no single superphosphate SSP applied), low fertility (LF; 125 kg SSP ha⁻¹ yr⁻¹), and high fertility (HF; 375 kg SSP ha⁻¹ yr⁻¹), since 1980. Slope classes: Low slope (LS; <12°, outer circle), medium slope (MS; 12–25°, middle circle), and high slope (HS; >25°, inner circle). Aspect locations: East (E; 35–155°), Southwest (SW; 155–275°), and Northwest (NW; 275–35°). These examples were set for the chance of detecting a significant difference (the power) at 80 % and we use an α -level of 0.05.

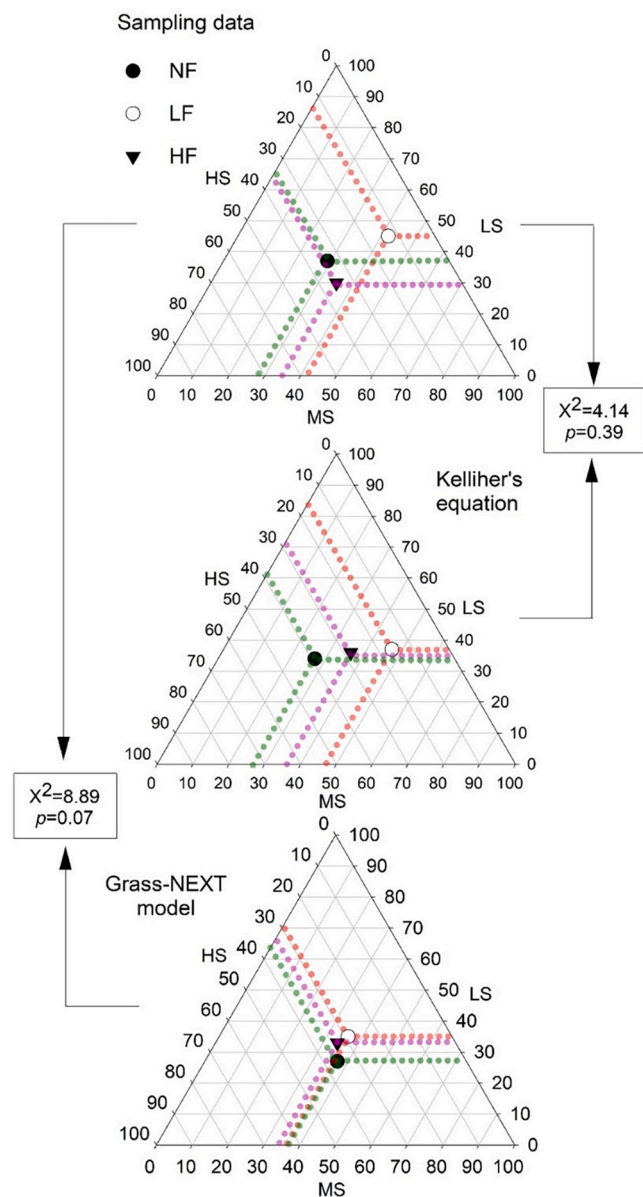


Fig. 7. Spatial allocation of soil samples required to detect significant differences between different methods to estimate changes in SOC stocks in a period of three years (contingency tables in Table S3 and S4, Supplementary Material). χ^2 : Chi square statistic. p -value < 0.05 means significant differences between the proportion of samples allocated by slope and farmlet. LS: low slopes, MS: medium slopes, HS: high slopes.

averages become difficult to understand when (i) the number of observations increases, and (ii) the rationale for weighting becomes subjective. We recognize that proportional and Neyman allocation methods can minimize the variance of estimators (Zhang et al., 2020). However, with a limited number of samples, proportional allocation may not adequately cover each stratum with multiple replicates and does not account for strata variance. In topographically complex landscapes with strata showing a coefficient of variation higher than 20 %, Neyman's allocation assumes strata are internally homogeneous. If this assumption is violated, its benefits are diminished (Latpate et al., 2021). Additionally, Neyman's allocation relies heavily on accurate prior information about stratum variances and sizes, making it complex, costly, and challenging in fieldwork (Mukherjee, 2023). Here, the obtained differences in mean and variability of SOC stocks (Figure S3 and S4) highlight the need for consistent sampling design, area coverage, and spatial

variability over time, beyond the selected stratified sampling allocation approach.

4.2. Representativeness of medium slopes for estimating SOC stocks in hill country pastures

A measure of SOC stocks to 300 mm on the medium slope was a reasonable estimate of the SOC stocks across all slopes (Fig. 4). This finding is supported by previous studies have reported that intermediate levels of P (Mackay and Lambert, 2011), SOC concentrations (Mackay et al., 2021b), earthworm abundance (Schon et al., 2022), soil moisture and microbial activity (Zhong et al., 2016) on medium slopes. Animal behaviour results in the transfer of nutrients in dung and urine from medium and high slopes to low slopes, resulting in significantly higher nitrogen (N) excreta on low slopes (Hoogendoorn et al., 2016). Consequently, low slopes have a higher N cycling rate, with annual excreta N loads two- and four-fold greater than on medium and high slopes, respectively, and a higher proportion of N leached. The SOC and soil N concentrations are governed by isometric stoichiometry (Cleveland and Liptzin, 2007), and macronutrient cycling in grazing systems is closely related to CO₂ fluxes and SOM turnover (Dungait et al., 2012). This topographic nutrient flux imbalance based on trials and biological evidence explains why medium slopes provide an intermediate SOC stock, serving as a good approximation for the sampled farmlets.

While full-depth soil sampling at 0–300 mm indicated that medium slopes are the most representative for SOC monitoring, our spatial analysis revealed statistically significant differences in SOC stocks in the deepest layer (150–300 mm) across slopes (Fig. 4 and Figure S3). The possible mechanism that may explain these differences between slopes in the deepest layer is a different rate of dissolved organic matter movement in soil solution, by varying mineral adsorption on clay and altering the incorporation and mixing of fertiliser material, plant litter and topsoil by earthworms (Mackay and Costall, 2016). Although the topsoil may have similar root biomass across different slopes, the subsoil can show variations in root penetration and organic matter input (Tian et al., 2020). In particular, LS with deeper root systems and additional organic matter from runoff tend to have increased SOC levels in the subsoil (Saggar et al., 1999). As a starting point for hill country pastures, our findings suggest that SOC stocks at 0–300 mm depth, or shallower samples (0–75 mm depth) on the MS position provide an adequate representation of SOC stocks across all three slope positions.

4.3. Shallow sampling performance to describe vertical distribution of SOC stocks

Depending on the objectives, production system, and time scale, we confirmed that estimating SOC stocks for the full profile (0–300 mm depth) can be effectively performed using shallower soil depths (0–75 mm) with faster and cheaper quantification techniques. The SOC stocks observed in 2003 and 2020 showed a strong, significant relationship with those estimated using equations calibrated from SOCd calculations across slopes and farmlets (Fig. 5). This confirms a consistent vertical distribution of SOC stocks over time, suggesting that these calibrated exponential equations can be used to fill data gaps and estimate SOC changes at a depth of 0–300 mm in long-term pastures. Furthermore, Kelliher et al. (2013) found that despite the varying properties of soils groups covered by pastures, total SOC storage could be estimated using a single regression slope. This method is unsuitable for farm systems with frequent crop rotation and soil tillage due to likely changes in the vertical distribution of SOC stocks. Ideally, soil sampling should cover the entire root zone to accurately report SOC stocks (Olson and Al-Kaisi, 2015). Therefore, it is crucial to appropriately stratify the soil layers and select the right regression model to accurately describe the SOC profile, considering variations in soil types, climate, and vegetation management (Ingram and Fernandes, 2001; Murphy et al., 2019).

4.4. Spatial and temporal SOC stocks modelling as an instrument for monitoring and reporting on SOC stocks in complex landscapes

The modelled results are of key relevance for planning soil sampling strategies and measuring SOC changes in topographically complex, long-term pastures (Figs. 6 and 7). While the modelled sites with high initial SOC stocks exhibited a gradual reduction of SOC stocks across slope classes and aspect locations, the sites with low initial SOC stocks showed no change or a slight gain (Fig. 6). Based on international studies conducted on long-term temperate grazing systems, a linear increase of SOC and soil N stocks would be expected in soils with low SOM levels or after crop rotations (Coonan et al., 2019). However, when soils are near C saturation covered by long-term pastures (e.g. HF farmlet), the large amounts of C and N inputs could be counterbalanced by an exacerbation of their losses or a significant reduction of inputs, maintaining or slightly declining SOC stocks (Bilotto et al., 2022). The lower simulated SOC changes in the 10-year period estimated for the HF farmlet (1–3 % change from initial SOC stocks, ~ 0.5 Mg C/ha) (Fig. 6) and the similar spatial and temporal variability compared to the NF and LF farmlets, indicate that the HF farmlet requires a more intensive sampling regime (>250 samples, Fig. 6 and Figure S3).

The modelling of alternative topographical configurations in the HF farmlet confirmed that the required number of samples to accurately represent to monitor SOC stocks in hill country pastures is sensitive to the proportion of slope and aspect. While the HF farmlet has shown a relatively spatial uniform mean SOC changes and variability, the NF and LF farmlets had tangible reductions in SOC stocks in the SW and E aspects and increases in SOC stocks in LS with NW and E aspects ($\sim 5\%$ change on initial SOC stocks), respectively. Consistent with our results, Mackay et al. (2021a) reported that the highest spatial variation in SOC stocks occurred in LS with NW aspects. Soils facing NW and E aspects would require a longer period to detect a significant MDD change of 3 % in total SOC stocks. The level of net primary production in the low P fertility farmlets (NF and LF) and modelled C inputs (litter and roots), might reveal significant changes in SOC after 7–15 years (Bilotto et al., 2022; Smith, 2004). But the exacerbated SOC losses in soils with high P fertiliser application (HF farmlet) and near C saturation suggests that even when increasing C inputs, the SOC accumulation rate will become asymptotic, delaying the detection of significant SOC changes to 25–45 years (Fig. 6).

4.5. Integrating spatiotemporal accounting methods and modelling approaches for reporting on SOC stocks

The ultimate focus of this study was to foster options for integrating SOC stock monitoring into routine soil nutrient fertility sampling (e.g., pH, Olsen P, K, Mg, S), and the incorporation the results into the farm plan and reporting process (Synge et al., 2013; Mackay et al., 2018; Australian Government, 2022). Even though only 30–50 % of farmers in developed countries like New Zealand, England, and Australia adopted routine soil sampling (Lobry de Bruyn, 2019; Rhymes et al., 2021), building monitoring on existing resource and nutrient management approaches, increases the likelihood of adoption and uptake. The proposed approaches allow farmers to expand their understanding about the vertical distribution and spatial variability of SOC stocks, in response to the demands by the shareholder, regulator or market.

Systematically, we recommend stratifying a defined area for SOC stock estimation into strata where SOC stocks and changes are relatively uniform to obtain precise estimates and reduce sampling costs. For benchmark sampling, collect samples randomly across the farm; for subsequent monitoring, collect samples in close proximity to the initial sites (Mudge et al., 2020). Aerial photos, paddock maps, and digital soil type and slope maps can aid in stratifying sampling areas. Here, we have identified slope classes (LS, MS, HS) and aspect locations (E, SW, NW) as key micro-topographical gradients (Mackay et al., 2021a). Beyond topographical features, grazing management plays a key role as the

selective use of hilly pastures by cattle and sheep significantly affects nutrient distribution, influencing landscape heterogeneity and the spatiotemporal variability of SOC stocks. (Jewell et al., 2007; Bilotto et al., 2022). The observed spatial distribution and modelled temporal variability of SOC stocks in topographically complex grazing landscapes suggest that a comprehensive approach, incorporating simple but context-specific methods, is essential for effective SOC monitoring. Collectively, it is important to note that results reported herein were obtained from i) one location, with distinct topographic features and a prominent soil type, and ii) a long-term permanent pasture (no cultivation nor disturbance, other than grazing sheep). Also to note, the Grass-NEXT model was largely developed with data obtained from the same location.

5. Conclusions

This study highlights the critical importance of tailored soil sampling strategies for accurately monitoring SOC stocks in topographically complex landscapes. Stratified sampling has proved to be more effective than random sampling by significantly reducing variance and enabling a more accurate detection of a 5 % deviation in SOC of the sample mean. For subsequent monitoring rounds the samples should be collected in fixed proximity, given that this design requires the least number of samples to detect changes in SOC stocks. Within the strata, medium slopes were identified as a representative stratum, making them a reliable target for estimating and monitoring SOC stocks in long-term pastures. As a faster, cost-effective method and viable alternative for filling data gaps in time series or addressing missing depths in some samples, our findings underscore the potential of using shallow soil sampling (0–75 mm) combined with calibrated exponential equations to estimate the vertical distribution of SOC stocks. However, its effectiveness varies with soil management practices and consistent land use over time as it is highly dependent on comprehensive stratification and prior full-depth sampling (0–300 mm).

The modelling results confirmed the sensitivity of SOC stock estimates to topographical variations, emphasizing the need for intensive sampling regimes in areas with high initial SOC stocks, near carbon saturation (e.g. long-term pastures and soil with high fertility in low slopes) and high spatial variability (facing NW and E aspects). These insights advocate for integrating SOC monitoring into routine farm planning and leveraging existing tools developed for local contexts to enhance precision and feasibility. Overall, this work provides pragmatic recommendations for SOC stock estimation and future research should continue refining these methods to support sustainable land management.

CRedit authorship contribution statement

Franco Bilotto: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Ronaldo Vibart:** Writing – review & editing, Writing – original draft, Investigation, Data curation, Conceptualization. **Alec Mackay:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Dongwen Luo:** Writing – review & editing, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Zachary Dewhurst:** Methodology, Investigation, Data curation. **Brian Devantier:** Methodology, Investigation, Data curation. **Emma Noakes:** Methodology, Investigation, Data curation. **Matthew Tom Harrison:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

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